

Sentence representations for semantic textual similarity: A systematic review[☆]

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ABSTRACT

In natural language processing (NLP), generating semantically-rich representations of sentences can improve performance on multiple tasks, such as question answering, duplicate detection, sentiment analysis, and machine translation. Recent approaches to NLP using machine learning can produce text representations that carry syntactic and semantic information. This article surveys recent works on generating sentence representations for semantic textual similarity tasks. We conduct our survey using a systematic literature review approach. We retrieve papers from several digital libraries and summarize their key techniques and findings. We propose a taxonomy to facilitate the understanding of the semantic textual similarity task on the sentence level. In our analysis, we describe the current state-of-the-art in sentence representation for semantic textual similarity and propose a guideline for working on this task.

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1. Introduction

The use of machine learning models to perform natural language processing (NLP) tasks requires a conversion of text into a numeric representation. Since machine learning models deal with the multiplication of an input by a series of weight sets, we need a numeric representation for the text that can capture its main characteristics. Research on NLP developed various representations to express different text features, such as syntactic and semantic information of specific text segments, such as words, sentences, and documents (Vaswani et al., 2017; Devlin et al., 2018; Reimers and Gurevych, 2019a).

Multiple tasks in NLP have sentences or entire documents as their main focus. Tasks such as Machine Translation, Question Answering, and Information Retrieval process sentences or multiple sentences to obtain the expected result. For these tasks, generating a semantically-rich representation of sentences directly impacts the final result of these tasks. Therefore, mapping texts to a high-dimensional space improves the results of NLP tasks when the values attributed to each piece of text can carry both syntactic and semantic information.

Multiple approaches for sentence representation were proposed in recent years, however, an analysis and evaluation is necessary to compare these approaches and understand which ones produce the best representations. Thus, in this work, we present a systematic literature review of sentence representation for semantic textual similarity (STS) tasks.

Our systematic literature review tries to fill the gap in the existing literature, bringing a broad review for sentence representation. Existing works focus on a particular subject, such as sentence representations or similarity measures. More specifically, there are works about universal sentence representations (Li et al., 2022), methods for measuring sentence similarity (Farouk, 2019a), the evolution of semantic similarity between texts (Chandrasekaran and Mago, 2021), semantic textual similarity (Zad et al., 2021) and semantic similarity of short texts (Han et al., 2021). In this work, we try to cover most of the aspects involving sentence representations in a single research. We follow a protocol review and focus on extracting information from existing approaches. This is necessary to identify patterns and to define the state-of-the-art for sentence representation in the semantic textual similarity task.

We conducted a study review of existing digital libraries to identify these approaches, their methodologies, datasets, and metrics. As a result, we obtained a series of papers about the topic from 2018 to February 2025. We propose a taxonomy to organize the task and a series of visualizations of the extracted data. We also conduct data analyses and propose a guideline for working on sentence-level semantic textual similarity. Our main contributions are:

- A systematic literature review of the state-of-the-art in sentence representation for semantic textual similarity;
- A general description of the works on sentence representation;
- A series of analyses over the extracted data containing the description of the best models and the obtained results on the STS datasets;
- A guideline for selecting models and methods for the semantic similarity between sentences.

This paper is organized into 6 sections. In Section 2.1, we describe the main concepts that concern this work. In Section 3, we introduce our research protocol for this systematic review. In Section 5, we perform an analysis over the publications obtained with this review. We discuss the statistics on the selected papers and provide a guideline for working on sentence representation for STS. We describe related work in Section 4 and compare the existing work against ours. Finally, in Section 6, we conclude the paper by describing the main points of this work and the results obtained.

2. Text representation and semantic similarity

In this section, we describe the two main concepts of this systematic review. First, we discuss text representation as a method to convert natural language into a vector structure that allows us to process text considering its syntax and semantics. Then, we define the concept of semantic similarity between texts.

2.1. Text representation

In order to process natural language, we need a representation that can capture the relevant features of a text. Throughout the years, developed approaches represent texts in different granularity, such as words, phrases, sentences, and documents. One of the most basic and common ways to represent a document (a sequence of words) is using a bag-of-words (BoW) (Zhang et al., 2010). With BoW, we convert a document into a sparse numerical vector of integers. Each position in the vector represents a single token, and the value of this position denotes the number of times it appears in the document. Consider a set of documents $D = \{d_1, d_2, \dots, d_n\}$, where d_x is a single document. Then, consider that each document is a set of words $d_x = \{w_{x1}, w_{x2}, \dots, w_{xm}\}$, where w_{xy} is a word with a subscript defining the document identifier x and its position in the document y . If we want to represent d_x using BoW, first, we need to compute the vocabulary for D , which in most applications represents all existing words in the set. In our example, the vocabulary is the following set: $V = \{w_{11}, w_{21}, \dots, w_{x1}, w_{x2}, \dots, w_{xm}\}$. The vocabulary size ($|V|$) often corresponds to the size of each document representation provided by BoW. Once we have the vocabulary, to represent d_x , we simply count the number of occurrences of each word in the document and update the final representation. Eq. (1) shows how BoW represents d_x , where n represents the number of occurrences of a word w_{pq} .

$$\text{BoW}(d_x) = [n_{w_{11}}^1, n_{w_{21}}^2, \dots, n_{w_{xm}}^{|V|}] \quad (1)$$

Similarly, the Term Frequency Inverse Document Frequency (TF-IDF) approach represents documents using the same logic as BoW, i.e., each document is a vector and the position represents a token. However, TF-IDF computes the value of each position of the vector based on the frequency of the token in both the represented document and the rest of the documents. This means that this representation takes into consideration not only the presence of a term in a document but also its presence over the others. Thus, the position value works as a weight to reflect how representative the specific term is for the document (Rajaraman and Ullman, 2011). Given the set of documents D and a document d_x , we can obtain the resulting value of each word in d_x with three steps. First, let w_{xy} be one of the words in d_x , we compute its frequency in the document. Eq. (2) shows how we can obtain the frequency, where freq is a function that counts the occurrences of a word in a document. The frequency of w_{xy} in d_x is divided by the sum of the frequency of all words in d_x . Once we have the frequency of w_{xy} for the document we aim to represent, we check its frequency in the other documents in D . Eq. (3) calculates the natural logarithm of the total number of documents in D ($|D|$) divided by the number of documents containing w_{xy} . Finally, we obtain the value for w_{xy} in d_x by multiplying TF and IDF , as shown in Eq. (4). The obtained value for w_{xy} is placed in the position referring to the word. Thus, the resulting vector from TF-IDF has the same size as a BoW, but with a weighting value instead of a counter.

$$TF(w_{xy}, d_x) = \frac{\text{freq}(w_{xy}, d_x)}{\sum_{w_{xi} \in d_x} \text{freq}(w_{xi}, d_x)} \quad (2)$$

$$IDF(w_{xy}, D) = \log \frac{|D|}{\sum_{d_i \in D} 1 |w_{xy} \in d_i|} \quad (3)$$

$$TF - IDF(w_{xy}, d_x, D) = TF(w_{xy}, d_x) * IDF(w_{xy}, D) \quad (4)$$

These approaches work with the concept of representing texts by their words only. This means that two documents are only similar when they share the same words. Thus, the position of words in the vector does not reflect their position in the document. Such feature can create an inconsistency when we compare sentences with the same set of words, but different word order. They can look the same, but the meaning can be the opposite.

More recently, new methods for representing text have been developed. They involve the creation of distributional representations for texts through the training of neural networks (Li et al., 2022). This new way of creating text representations consists of using a neural network to learn the relation between words. It is possible to train a neural network to predict a word given the surrounding words as input. The resulting internal representation from this training is used to represent the specific word. Such a new method can preserve both syntactic and semantic information about the word. Mikolov et al. (2013) first introduced the concept of distributed representations. In their work, they develop two models: Continuous Bag-of-Words (CBOW) and Continuous Skip-Gram Model. These models generate representations of words using the context in which these words are placed. They train CBOW to predict the current word of a text using four words before and four words after the target word as input. On the other hand, they train the Continuous Skip-Gram Model to do the opposite, i.e., given a word as input, predict four words before and four after the sentence. Fig. 1 illustrates how to train both models. As we can see, W_i represents a word in a sentence and i indicates its relative position regarding its neighbors. In CBOW, W_0 is the expected output, and its surrounding words ($W_{-2}, W_{-1}, W_{+1}, W_{+2}$) are the input. This makes CBOW faster when compared to Skip-Gram and focused on the local context of words. Skip-gram, on the other hand, focuses more on rare words, preserving the global context. The representation of each W_i is a sparse vector where all positions are 0 except the one representing the word. The input vectors are projected into the hidden layer and processed to generate the output word.

Since the publication from Mikolov et al. multiple researchers developed other approaches. GloVe (Pennington et al., 2014) improves on Mikolov et al. to provide a richer representation of words. The authors try to combine both local and global contexts of words. New approaches started to use a similar methodology to generate representations for entire sentences and documents (Pagliardini et al., 2018; Moghadas and Zhuang, 2020). These approaches were able to represent a sentence in a dense vector that preserves most of its characteristics.

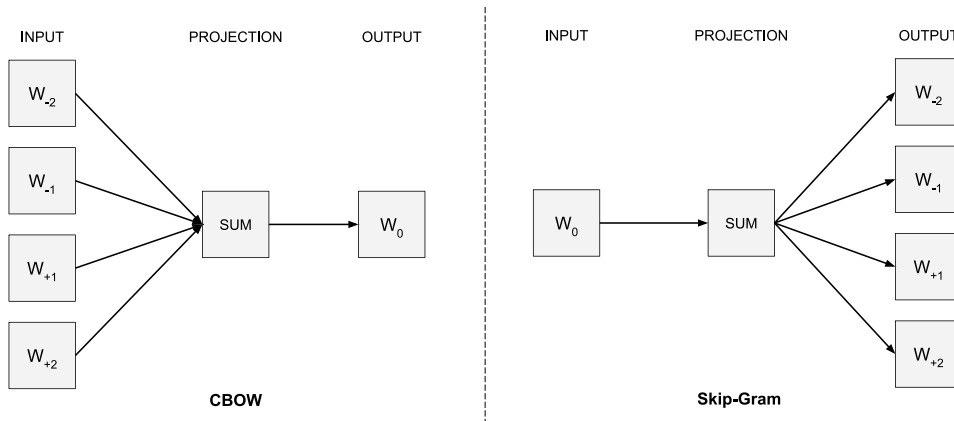


Fig. 1. CBOW and Skip-Gram models.

More recently, Vaswani et al. (2017) introduces the basis of current Transformer models. This new model consists of an encoder-decoder architecture that is capable of generating rich representations from texts by using a stack of attention layers over a positional encoder representation that preserves the position of words in a sentence. Using the attention layers, the model can connect the influence that one word has on another and express it in the representation. Transformer models helped originate other models such as BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2018). BERT is a modification of the encoder part of the Transformer model. As its name suggests, it is used to generate text representations based on the Transformer structure. The main difference from the previous approaches is that the representation of a word is unique to its specific concepts, *i.e.*, words with more than one meaning will be represented differently according to the context. Thus, words like “bank” that can refer to either “a financial establishment” or “the land alongside a river”, will have more than one representation depending on the context. From BERT, new approaches emerged, such as RoBERTa (Y. Liu et al., 2019), ALBERT (Lan et al., 2019), and BART (Lewis et al., 2019). All these approaches improve some aspects of BERT in the generation of text representations.

2.2. Semantic similarity between texts

The semantic similarity represents a degree of how connected two pieces of text are in terms of meaning (P. and Shaji, 2019). We have different similarity metrics between texts that can bring us different perspectives. Texts can be similar in how many words they share, by their size, syntactically, and semantically. When dealing with the semantics of two texts, we must consider their meaning only. Two texts can have the same words but different meanings, such as in the example below.

- 1 Lucas studies biology, not geography.
- 2 Lucas studies geography, not biology.

This example, although simple, shows that just that the mere presence of words in a sentence does not carry enough information to convey the meaning of the sentence. By contrast, two sentences can have with no words shared between them, and yet convey the same meaning.

- 3 It will rain today.
- 4 Showers are expected throughout the day.

The sentences above are semantically similar, *i.e.*, they describe the same situation. However, they do not share a single word, which can mislead traditional methods of measuring semantic similarity.

2.2.1. Similarity metrics

In order to measure the similarity between distributional representations of text, it is common to calculate the distance between them. Since we are comparing vector representations in the same space, we can use existing distance metrics, such as *Cosine*, *Manhattan*, and *Euclidean*.

The Cosine similarity calculates the cosine of the angle between two vectors. For instance, consider two vectors $V_1 = [0.8, -1.5, 0.5, -0.4]$ and $V_2 = [1, 9, -0.5, -0.2, -1.6]$ in four-dimensional space. To obtain the cosine similarity between them, we need to compute the dot product of the vectors and divide it by the product of the vectors' length. Eq. (5) shows the similarity computation between V_1 and V_2 . As we can see, the resulting similarity of both vectors is 0.61. To obtain the cosine distance, we only need to subtract cosine similarity from 1. Eq. (6) illustrates the computation of the cosine distance for both V_1 and V_2 , which results in 0.39.

$$\cos(V_1, V_2) = \frac{V_1 \cdot V_2}{\|V_1\| \|V_2\|} = \frac{2.81}{4.59} = 0.61 \quad (5)$$

$$\cos_dist(V_1, V_2) = 1 - \cos(V_1, V_2) = 0.39 \quad (6)$$

The second distance metric used for computing the similarity is the Manhattan distance. To compute the Manhattan distance, we compare two vectors to obtain the sum between their absolute differences. Eq. (7) computes the Manhattan distance for vectors V_1 and V_2 . As we can see, the resulting distance between both vectors is 4.0. What changes from the Cosine similarity is that while the Cosine ranges between -1 and 1 , the Manhattan distance returns a positive value that will be as large as the relative distance between the vectors.

$$M(V_1, V_2) = |V_1 - V_2| = \sum_{i=0}^4 |V_1^i - V_2^i| = 4.0 \quad (7)$$

Finally, the Euclidean distance computes the line that connects two points. The equation is similar to the Manhattan distance. However, Manhattan distance works better on high-dimensional vectors. Eq. (8) shows the computation of the distance between V_1 and V_2 .

$$euc(V_1, V_2) = \sqrt{\sum_{i=0}^N (V_1^i - V_2^i)^2} = 2.03 \quad (8)$$

2.2.2. Evaluation metrics

Working with Semantic Textual Similarity implies that we have to compare the output from models with the similarity scores annotated by humans. Thus, we need metrics to indicate how close the model prediction is to the gold standard. Similarity scores annotated by humans are often an average similarity value between multiple annotations or a single class, indicating whether two sentences are similar (1) or not (0). For the former, where the result is often a floating number, the most used metrics are Pearson and Spearman correlation. For the latter, metrics such as accuracy, prediction, recall, and F-measure are used.

When we have the similarity annotation from the average of multiple human annotators, the resulting value is often a float number. For instance, consider s_i as a sentence and P as a list of three sentence pairs: $P = [(s_1, s_2), (s_2, s_3), (s_3, s_1)]$. Then, consider a_i as an average similarity from human annotators and HA as a list of the average similarities for each pair in P : $HA = [a_1, a_2, a_3]$. Finally, consider the output of a model M for each pair in P as $M_{out} = [m_1, m_2, m_3]$. We calculate the correlation between HA and M_{out} using either (or both) Pearson and Spearman correlation. The Pearson correlation coefficient is a correlation metric that calculates the linear association between two sets of data in terms of strength and direction. Eq. (9) shows how we can calculate the Pearson correlation for V_1 and V_2 . First, we need to obtain the covariance between V_1 and V_2 , where $\bar{V}_1$ and $\bar{V}_2$ represents the mean of V_1 and V_2 , respectively. As a result, we obtain a value between -1 and 1 , where 1 indicates a positive relationship between the data, -1 is a negative relationship, and 0 indicates no relationship. In the context of evaluating the models' output, the Pearson correlation shows how close the models' predictions are to the human annotations.

$$\rho(V_1, V_2) = \frac{cov(V_1, V_2)}{\sigma V_1 * \sigma V_2} = \frac{\frac{\sum_{i=0}^N (V_1^i - \bar{V}_1)(V_2^i - \bar{V}_2)}{N-1}}{\sqrt{\sum_{i=0}^N V_1^i - \bar{V}_1} * \sqrt{\sum_{i=0}^N V_2^i - \bar{V}_2}} \quad (9)$$

The Spearman correlation is a different view of the Pearson correlation. Spearman ranks the sets of data and applies Pearson correlation over such ranking. The metric works on the monotonic association between data sets. Eq. (10) shows the calculation of the Spearman correlation between V_1 and V_2 , where R means the ranking of the elements in each set.

$$r_s(V_1, V_2) = \frac{cov(R(V_1), R(V_2))}{\sigma R(V_1) * \sigma R(V_2)} \quad (10)$$

3. Research methodology

This article surveys existing approaches to sentence representation for the semantic textual similarity task through a systematic literature review. The systematic review is a research method that strictly follows a pre-defined methodology based on research questions (Kitchenham and Charters, 2007). The main goal is to find most of the available literature about a specific topic to provide a synthesis and analysis of such literature. Fig. 2 illustrates the research protocol we use to conduct this systematic review (Kitchenham and Charters, 2007). In the following sections, we detail every step in the protocol.

Need for a systematic review. Converting raw natural language text into distributed representations provides multiple benefits in using text-related processes in machine learning and artificial intelligence in general. Natural language is often ambiguous and needs interpretation for its complete understanding by computers. We need rich representations that carry syntactic and semantic information when working on natural language processing tasks. Through these representations, we can identify semantic similar elements. Semantic similar texts can often have different words but similar meanings. Clustering texts and removing duplicates are two of the multiple tasks that can be benefited from representations carrying semantic information. Recent advances in models for text representations, such as Transformer (Vaswani et al., 2017), BERT (Devlin et al., 2018), and GPT versions (Brown et al., 2020) improved significantly the way we obtain richer representations. From these initial models, new ones develop modifications that obtain state-of-the-art results in multiple NLP tasks. The increasing number of new approaches makes it necessary to perform a systematic review that can list, analyze, and compare them. A systematic review can shed light on the different existing methods for reaching meaningful text representations. From the results of a systematic review, one can select the best suitable method for a specific application scenario. Thus, a systematic review can provide the research community with an organized view of this topic.

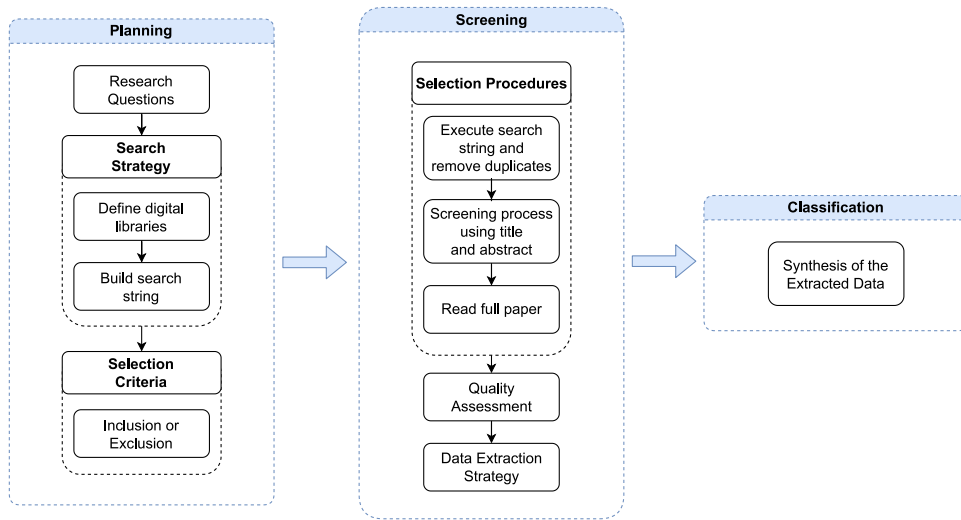


Fig. 2. Research methodology.

Table 1

Databases used to obtain the studies for this systematic review.

Database	URL
ACL Anthology	http://aclanthology.org
ACM Digital Libraries	https://dl.acm.org/
IEEE Xplore	https://ieeexplore.ieee.org/Xplore/home.jsp
Scopus	http://www.scopus.com

Systematic review goals. In this systematic review, we aim to investigate the existing literature about sentence representation for the semantic textual similarity task. Although we refer to “sentences” throughout this work, we aim to obtain solutions that encapsulate entire documents also, *i.e.*, texts longer than a single word and that can contain more than a single sentence. Existing surveys (Farouk, 2019b; Zad et al., 2021) show a small or limited range of papers about the subject. With the consistent improvements in the area, we aim to approach the subject by discussing more recent advances (from 2018 to February 2025) and applications using sentence embeddings being employed for semantic similarity computation. We consider publications from a period that starts in 2018 due to the fact that the Transformers architecture (Vaswani et al., 2017) was published in 2017, and such architecture became the basis of multiple recent approaches for sentence representation (Chandrasekaran and Mago, 2021). To the best of our knowledge, there is no existing systematic literature review about sentence embeddings for semantic similarity. Thus, this work will be the first on the subject.

Research questions. We guide our systematic review with two research questions. They indicate our research objectives in terms of what we aim to answer at the end of this systematic review.

RQ1 Which semantic representations for sentences have been proposed and in what contexts?

RQ2 Which techniques for semantic representation can be used for semantic similarity computation?

RQ1 question aims to discover the multiple ways we can represent sentences. Representations can extract different features of sentence carrying, as a result, have an impact on semantic similarity identification. On the other hand, **RQ2** focuses directly on the sentence representations already being used for obtaining the semantic similarity of sentences.

Search strategy. We divide our search strategy into two main steps. First, we define the digital libraries to source publications for our review. Then, we built a search string to fetch the publications within the expected context. Table 1 lists the databases we used to retrieve the publications. We selected the digital libraries based on the potential to find publications from the specific field of natural language processing. The final list consists of four digital libraries.

Our second step to identify the publications for the systematic review is the definition of a search string. The search string considers three aspects: population, intervention, and outcome. The population is the main focus of the review. For this work, our population is *text* and *sentence*, as we aim to identify representations of them for the purpose of semantic similarity. The intervention refers to the methods and techniques used to modify the population. In this case, the intervention we envision is the *embedding* transformation of texts or, in other words, the *representation* of them. Finally, the outcome is the expected result of applying the intervention over the population. The expected result of applying such intervention is to obtain the *semantic similarity* between sentences/texts. Table 2 shows the elements that constitute our search string. We use the resulting search string in each selected database to obtain the papers.

Table 2
Search string elements.

Population	“text*” OR “sentence*” OR “raw text”
Intervention	“embedding” OR “representation” OR “vector space”
Outcome	“semantic similarity” OR “semantic association”

Selection criteria. In order to select the retrieved papers, we defined a set of criteria. They are divided into inclusion and exclusion criteria.

- **Inclusion Criteria:** Our inclusion criteria draw the limits to what we expect from a paper. So, we include any recent study that describes an approach to represent sentences/texts as vector spaces/embeddings, and that evaluates such a representation on a semantic textual similarity task;
- **Exclusion Criteria:** Our exclusion criteria ensure that we select only papers from a specific period, language, and methodology. In total, we defined six exclusion criteria: (1) duplicated papers from a different source; (2) Publications focusing on word embeddings; (3) Publications that do not compare their approach against existing literature; (4) Publications written in a language other than English; (5) Papers published before 2018; and (6) Publications that provide sentence representations in a language different from English.

Selection procedures. Given our selection criteria, we divide our selection process into three main stages. In the first stage, we execute the search string in the digital libraries. Then, we remove duplicates through DOI comparison. In the second stage, two researchers conduct the screening process, where the title and the abstract are evaluated and classified as included, excluded or uncertain. When there is uncertainty or disagreement, the paper should be discussed with a third researcher. Finally, in the third stage, two researchers read the full article and verify whether it fits all inclusion and exclusion criteria. Similarly to the second stage, in case of uncertainty or disagreement, a third researcher evaluates the article.

Quality assessment. In the third stage of the selection process, the researchers perform a quality assessment of the article. The assessment consists of seven questions that must be answered by the content of the article. The questions are: (1) Is it clear what dataset is being used to train/obtain sentence embeddings?; (2) Do the authors have a definition of the problem they are trying to solve with sentence embeddings?; (3) Do the authors propose a novel study?; (4) Are the evaluation metrics well-defined?; (5) Do the authors compare their results against existing work?; (6) Is there a semantic textual similarity evaluation of the representation?; and (7) Does the work deal with sentences/texts?

An article can be excluded if the answer to one of these questions is “no”. These assessment questions allow us to perform a more strict selection of the articles beyond the selection criteria. As a result, we obtain papers more aligned with our research questions.

Data extraction strategy. During the third stage of our selection process, the researchers extract information that we will use to summarize the studies and perform analyses. We have the following list of the information extracted from each paper: title; year of publication; affiliation; document type; country/location; training dataset; embedding approach; training model; evaluation metrics; results; evaluation dataset; and supervised/unsupervised approach.

Here, we consider that an approach to measure the semantic similarity between sentences consists of an Embedding Approach and a Training Model. This is not true for all approaches but serves to generalize a structure for extraction. The “Embedding Approach” represents the method that converts text into a distributional representation. On the other hand, the “Training Model” indicates the model used for computing the semantic similarity between text representations. Many approaches add or modify the structure of a pre-defined model (such as BERT and Transformer), for the “Embedding Approach” or “Training Model”, giving the new structure a specific name. In order to simplify the information in these fields, we use the name of the pre-defined model, as we can, later, cluster them based on this information. Sometimes, “Embedding Approach” and “Training Model” are the same given the approach structure, and we simply duplicate the information for both fields.

Synthesis of the extracted data. Finally, to define the expected analysis of the data, we propose a series of synthesis methods. First, we provide statistical information about the extracted papers. Then, we use the tabulated data, resulting from the Data Extraction phase, to cross information and obtain insights. The number of publications in each major category will be counted. We tabulate the results separately, and we derive graphs from them. We create a taxonomy to structure the different stages to obtain sentence representations that can be used for semantic similarity purposes.

Obtained publications. We obtained the publications used in this systematic review using our defined three-stage selection procedures. In the first stage, using our search string in digital libraries, we obtained 1708 papers. In the second stage, we discarded 1405 publications through the title and abstract analysis, leaving us with 209 publications selected for the next stage. Finally, in the third stage, we read the entire papers considering our Quality Assessment list and selected 119. An extra search was done after our selection to identify papers not selected by our systematic review. In the process, we identified three other papers to add to the final list of selected papers. Fig. 3 illustrates the three stages of paper selection, and Table 3 summarizes the number of selected papers in each stage, separated by the digital library. As we can see, IEEE Xplore and ACL were the major sources of papers for this systematic review. Together, they sum up more than 84% of the selected papers. The complete list of all reviewed papers is available on Table 7.

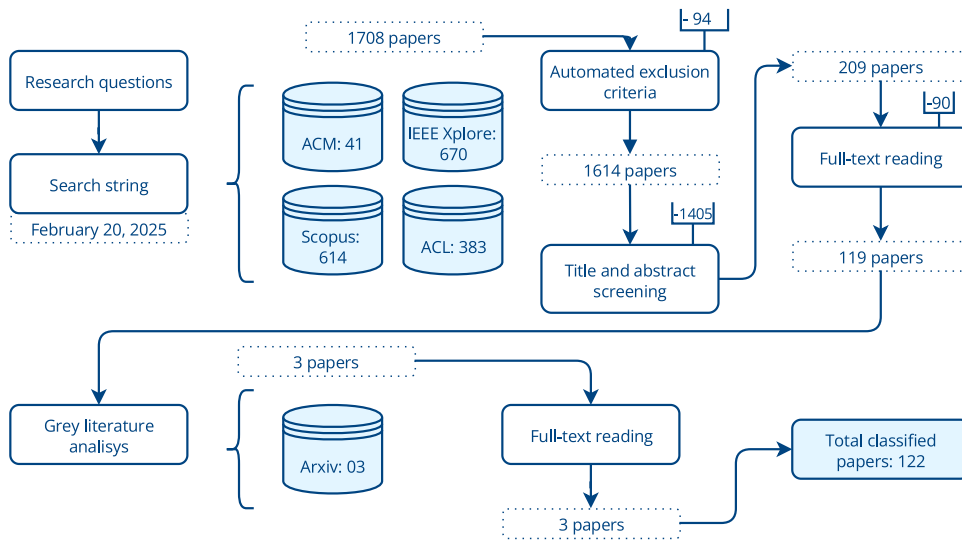


Fig. 3. Systematic review: Paper selection stages.

Table 3
Number of papers extracted from each digital library.

Digital library	First stage	Second stage	Third stage
ACL Anthology	383	61	44
ACM Digital Libraries	41	7	4
IEEE Xplore	670	101	57
Scopus	614	40	14
Grey Literature	–	–	3
Total	1708	209	122

4. Related work

In this work, we cover two main concepts: (1) sentence representation; and (2) semantic similarity. Existing surveys describe these concepts separately, and a few bring them together.

Li et al. (2022) provides an overview of methods to create universal sentence representations. They divide existing methods into four categories according to how the representation is created. Then, using downstream NLP tasks, they evaluate the performance of each type of representation. The authors select existing methods for each defined category and describe their training process. Among the downstream tasks presented, they evaluate the representations regarding their semantic similarity using well-known datasets. The results show that BERT-related approaches obtain the best results, while USE-DAN (Cer et al., 2018) (Universal Sentence Encoder for English) is the only method not using BERT that obtains competitive results.

Farouk (2019a) presents a survey about methods to measure sentence similarity. The author divides the similarity calculation into three categories: Word-based, Structured based, and Vector based. In order to evaluate the different metrics, the author selected a set of datasets, such as STS-12-17, STS-Benchmark, and the Microsoft Research Paraphrase Corpus. The results suggest that we can obtain better similarity measurements by merging different metrics types.

Chandrasekaran and Mago (2021) propose a survey to trace the evolution of semantic similarity between texts. They list proposed approaches used to measure the similarity over time and the different datasets for the task. The authors divide the techniques into four categories: knowledge-based, corpus-based, deep neural network-based, and hybrid methods. For each category, they detail the approaches and provide an analysis of the method.

Zad et al. (2021) present a survey about deep learning models used for measuring the semantic similarity of texts. Focusing on deep learning approaches, the authors describe the existing datasets and models usually employed for the task of semantic textual similarity. In the second part of the paper, they describe existing approaches specifically for generating sentence representations.

Han et al. (2021) proposed a survey focusing on the semantic similarity of short texts. The survey proposes a taxonomy of semantic similarity measures, divided into deep learning and non-deep learning measures. The authors also provide a division based on semantic similarity applications, datasets, and metrics used in the papers. In addition, a detailed comparison between the models using Mean Squared Error, Accuracy, and F1-Score metrics on selected datasets: MSRP, SICK, STS Benchmark, and Quora. Bali et al. (2024) proposes a review to check the process of information retrieval and text processing, that outperforms the ontology-based methods. In the semantic similarity synthesis, the authors revised only five papers. The results show that cosine and Jaccard similarity are the most used, and the papers focus on BERT to generate the representations. The main differences between existing

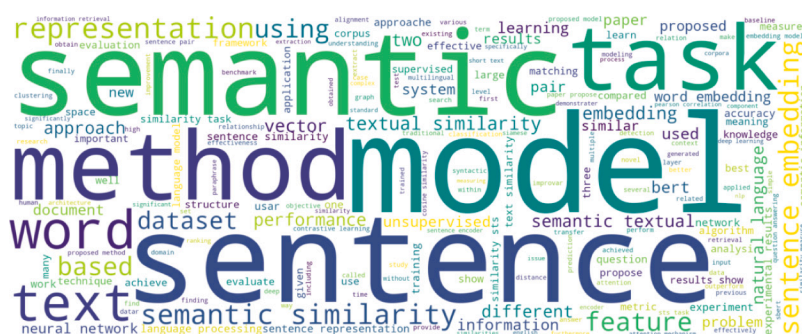


Fig. 4. Most frequent terms in the abstract of the selected papers.

work and ours are two. First, we propose a systematic review, which utilizes a detailed methodology for selecting existing studies from a specific topic. Thus, our goal here is to describe and extract information from the selected studies. The existing literature is focused on surveys that bring information about the research on a particular subject. On the other hand, a systematic review analyzes published works aiming to extract information from them, such as patterns and guidelines for working on the defined scenario.

The second difference between existing work and ours is the subject selected. In this work, we decided to focus on sentence representations for the semantic textual similarity task. Existing work describes either sentence representations or similarity measures. Describing these two topics together allowed us to connect information resulting in more specific approaches to the task.

5. Findings

In this section, we discuss the insights and analyses derived from our review. They include statistics about the selected papers, a taxonomy from the results, an analysis of the extracted data, and some guidelines to work on this topic.

5.1. General statistics

From the selected papers, we analyze the most frequent terms in their abstracts. We preprocess the abstracts by removing stop-words, converting terms to lowercase, and lemmatizing them. We used Spacy (Honnibal and Montani, 2017) for the preprocessing. Then, we generate a word cloud using the *word cloud* python library.¹

Fig. 4 shows the resulting word cloud of the abstract terms, where the size indicates the frequency. As expected, the most frequent terms are those also present in our search string. Some variations to the terms are also visible, such as “textual similarity” and “semantic textual”. Interestingly, tasks that benefit from the semantic similarity in texts also appear, such as “question answering” and “information retrieval”. Finally, the presence of “bert” indicates a tendency to the usage of a such model to represent and identify similarities between texts.

The first overview of the extracted data concerns the embeddings and models used. As an initial analysis, we counted the “Embedding Approach” described in papers to discover what are the most used ones. Fig. 5 displays the approach used to obtain the text representation in the x axis and the number of times it is used in the y axis. As we can see, BERT (Devlin et al., 2018) is the most used embedding approach among the selected papers. It is followed by Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), which indicates that many approaches work with word-level representations to create sentence-level ones.

As an initial understanding of the work on sentence representation for semantic similarity, we separated the “Embedding Approach” from the “Training Model Base”. The “Embedding Approach” is responsible for converting the text into a dense vector representation, while the “Training Model Base” receives the representations to compute the similarity between texts. Thus, similarly to the “Embedding Approach” visualization, we plot the most frequent models for obtaining the similarity value. Fig. 6 shows the model names and the sum of their usage among the selected papers. Again, BERT is the most used approach. This indicates that BERT is used to both generate the text representation and compute the semantic similarity between texts. In what follows, we have Custom, RoBERTa, Statistical Model, Transformer, Word2Vec and SBERT. We can also see some deep learning models, such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and Support Vector Machines (SVM). By “Statistical Model”, in the list of most used models, we mean those approaches that do not use conventional models, *i.e.*, well-known models. These approaches consist of a statistical model that follows a particular architecture.

¹ <https://amueller.github.io/word-cloud>

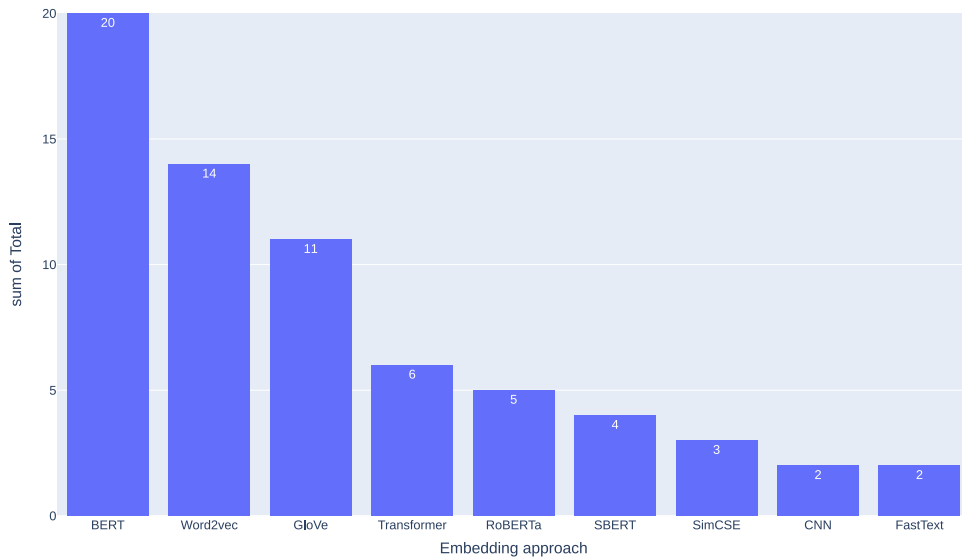


Fig. 5. Embedding approach.

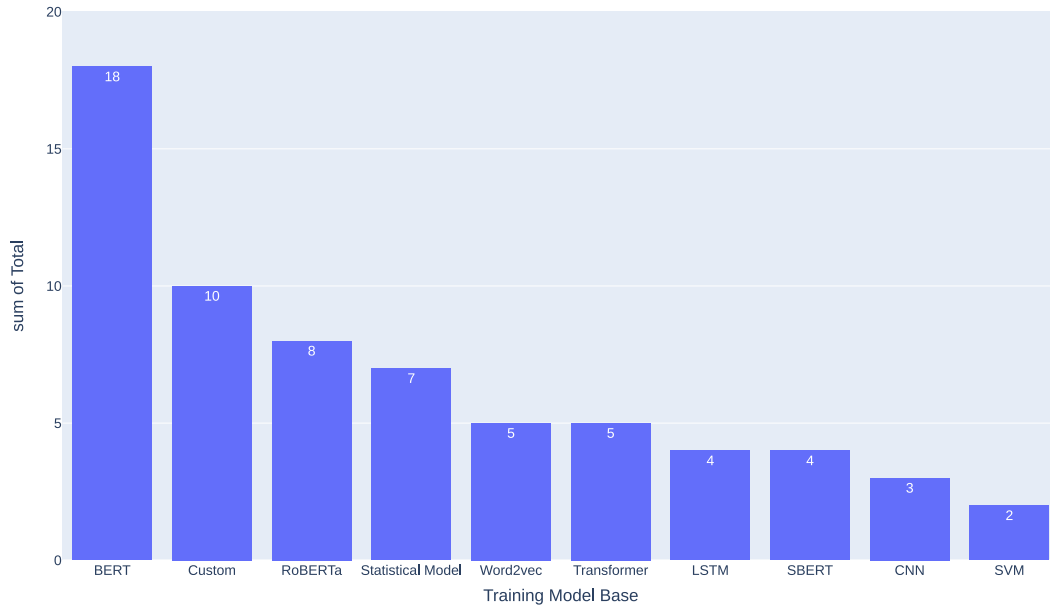


Fig. 6. Models used for computing the semantic similarity between sentences in our selected papers.

5.2. Taxonomy

As a summary of our work, we created a taxonomy that brings a general visualization of the extracted data. Our idea is to provide a guideline for those who will work on this task about what is more commonly used. Fig. 7 illustrates our taxonomy. It has at its root the use of sentence representation for STS (Semantic Textual Similarity). The main topic is subdivided into four subtopics: *Sentence Representation*; *Training Model*; *Evaluation Metrics*; and *Evaluation Datasets*. Each subtopic corresponds to one of the steps to work in the main topic. In *Sentence Representation*, we have types of methods for converting text into distributed representations. We separate them according to the focus of the representation. *Syntactic* lists two approaches that consider a text as a sparse vector that represents the presence of words in the text. The two approaches are BoW and TF-IDF, both explained in Section 2. *Learning Models* consists of approaches that generate text representation from existing multipurpose training models. This list has well-known models, such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory. Although we could merge both *Learning Models* and *Semantic*, we separate them because in *Semantic* we list only methods that were designed

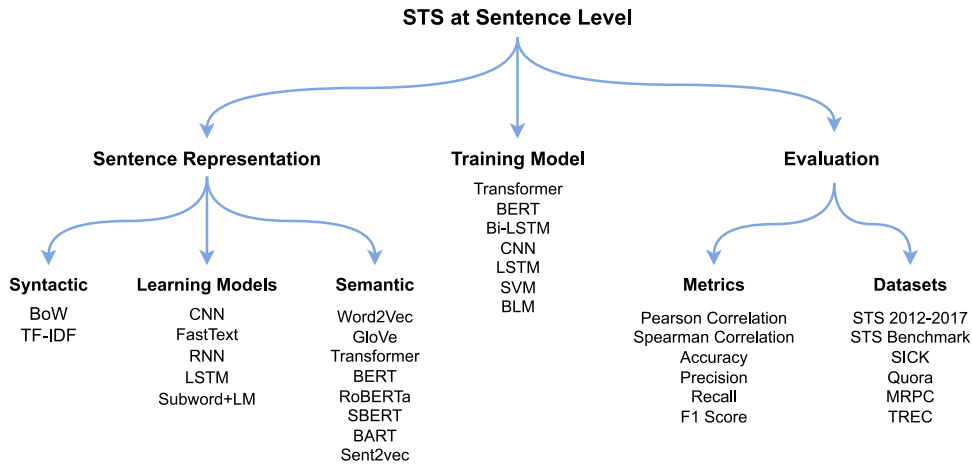


Fig. 7. Taxonomy of Semantic Textual Similarity at sentence-level.

to generate text representations. These methods generate dense vectors from texts, representing both their syntactic and semantics of them.

Training Model lists the models used for similarity computation. In this case, we have some of the models in the *Sentence Representation*, but also others used specifically for the task. Among them, we have Transformer, BERT, and SVM. Once we know what we can use to generate representations and which models use to train to calculate similarity, our next subtopic is the *Evaluation Metrics*. This subtopic lists the metrics usually applied to evaluate approaches. The most common ones are Pearson and Spearman correlation. These metrics show the correlation between the similarity scores between two sentences produced by the *Training Model* and the similarity provided by humans. In the case of a classification task, metrics such as accuracy, precision, recall, and F-Measure are also used. Finally, *Evaluation Datasets* lists the most common datasets used for semantic similarity evaluation. Most of them are part of the International Workshop on Semantic Evaluation.² In this event, humans annotate new datasets containing pairs of sentences and a semantic similarity score. STS 2012–2017 (Cer et al., 2017; Agirre et al., 2016, 2015, 2014, 2013, 2012) are six of the most used datasets from SemEval. STS Benchmark (Cer et al., 2017) consists of sentence pairs from STS 2012–2017.

5.3. Extracted data

In this section, we analyze and discuss some of the extracted data. First, we cross the information to obtain the relation between different fields. Initially, we wanted to understand how the “Embedding Approach” changed over the years. Fig. 8 shows the most frequent embedding approaches on the left side and the years (2018–2024) on the right side. As we can see, Transformer (Vaswani et al., 2017), BERT (Devlin et al., 2018) and Word2Vec (Mikolov et al., 2013) are the models that appear five different years. The year 2021 has the most connections with embedding models, which also demonstrates that we have an incoming significant change of approaches. In 2021 and 2022, we notice a tendency to use more recent models based on Transformer and BERT, while Word2Vec and GloVe were more predominant in previous years. Approaches like SimCSE (Gao et al., 2021), RoBERTa (Y. Liu et al., 2019) and SBERT (Reimers and Gurevych, 2019b) are more recent, emerging from 2021 to 2024. However, this is the analysis of our 122 selected papers, which does not completely reflect reality, only the representation of our selection.

In our analysis, we consider approaches as supervised and unsupervised. A supervised approach computes the similarity of two sentences using a trained model. This training consists of having pairs of sentences as input and a similarity score as output. In contrast, an unsupervised approach has a model that computes the representation of two sentences and compares them using a distance metric. Thus, the approach was not trained to measure the semantic similarity, but rather to generate meaningful representations that can be used for such computation. Considering this, we cross the information about supervised/unsupervised approaches with the Training Models. Fig. 9 shows three different scenarios. In green, we have models identified in supervised approaches. In red, we have models that use an unsupervised approach. Finally, in blue, we have models used in both supervised and unsupervised approaches. The size of the model block in each set indicates its frequency in comparison with the total number of elements. As the figure shows, some of the most used models are also those in both types of training, such as BERT, Transformer, and Word2Vec. For the unsupervised set, we can see that Custom is the most frequent. We defined “Custom” as those approaches that propose a specific method that does not rely on existing models, neither is a statistical model only. This indicates that for the unsupervised approaches, the authors tend to either use one of the well-established models or create a custom one that combines more techniques.

² <https://semeval.github.io/>

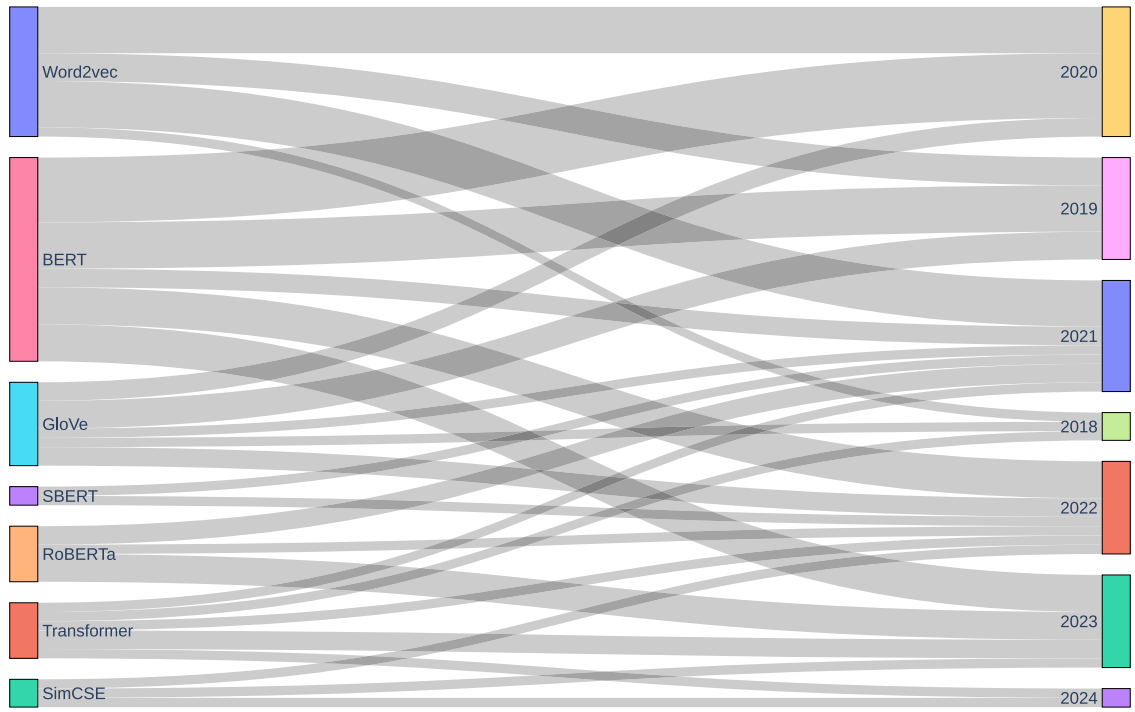


Fig. 8. Embedding approach over the years.

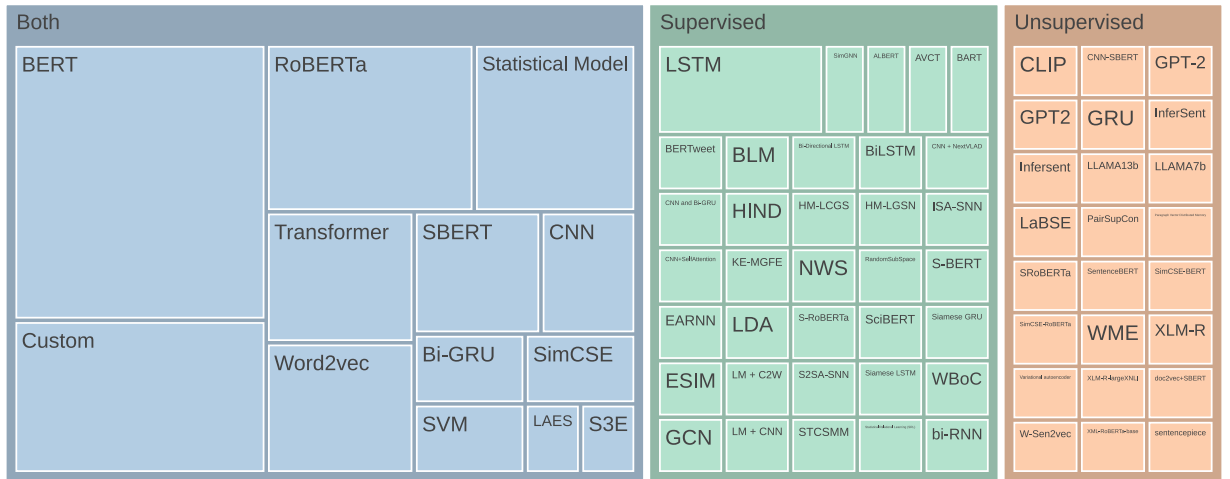


Fig. 9. Models × (Supervised/Unsupervised).

5.4. Models evaluation

In this section, we describe some of our findings when we compare the performance of the approaches in different datasets. Fig. 10 shows the best results obtained by the approaches in the SemEval datasets (STS 12–17, STS Benchmark, and SICK-R). The papers provide both Pearson and Spearman correlation as the metric for evaluation. We consider the best results of both metrics in each dataset. The results show that the majority of the best results come from approaches that use as basis SBERT (Reimers and Gurevych, 2019a). The paper from Choi et al. (2021), identified as CNN-SBERT in the figure, obtained the best result in five out of eight datasets selected. Their approach involves the use of both CNN and SBERT. On the other hand, the paper from Jiang et al. (2019) obtained the highest value for both Pearson and Spearman in the STS Benchmark, which comprises parts of the STS datasets from previous years. They use a modification of RoBERTa (Y. Liu et al., 2019) to create their approach. For the STS 2016, the best



Fig. 10. Best results for Pearson and Spearman correlation in each dataset.

result using Spearman correlation comes from a paper from [F. Liu et al. \(2021\)](#). While in STS 2017, the paper from [Ham and Kim \(2021\)](#) obtained the best result.

The paper from [Choi et al. \(2021\)](#) that obtained most of the best results in the selected datasets (STS 2013–2015; STS 2012 and 2016 Pearson only) introduced the CNN-SBERT model. The authors propose three modifications to SBERT ([Reimers and Gurevych, 2019a](#)). The first one replaces BERT in SBERT by ALBERT ([Lan et al., 2019](#)), creating the SALBERT model. Their second modification is the addition of a Convolutional Neural Network (CNN) as sentence embedding to SBERT, creating the CNN-SBERT. Finally, they combine the CNN and the first modification (SALBERT), creating the CNN-SALBERT. For the STS task, they fine-tune the model using the training data from the STS Benchmark. The results on the STS Benchmark show that the best version among the tested models is the CNN-SBERT with 85.91 on Spearman correlation. In the evaluation of STS 2012–2016 and STS Benchmark, the best average of Spearman correlation was on the original SBERT version. However, individually, CNN-SBERT is better on three out of six datasets (STS13, STS16, and STS Benchmark).

For the RoBERTa-based approach, [Jiang et al. \(2019\)](#) propose the SMART (SMoothness inducing Adversarial Regularization and BRegman pRoximal point OTimization) framework, which focuses on a fine-tuning method for pre-trained models. Their method regularizes the process of fine-tuning new data by modifying the loss function. They add hyperparameters to the function to control the model complexity in the fine-tuning phase. The framework was tested using RoBERTa as the model base, and as a result, on the STS Benchmark, the Spearman Correlation was 92.5, and the Pearson Correlation was 92.9.

[F. Liu et al. \(2021\)](#) proposed the approach that obtained the best Spearman correlation result in the STS 2016. The authors introduce the TRANS-ENCODER, an unsupervised approach to sentence-pair modeling, using self and mutual distillations. They create a bi-encoder based on the SimCSE ([Gao et al., 2021](#)) architecture. SimCSE is an unsupervised contrastive learning framework. Using RoBERTa-base model as their pre-trained language model, they create a cross-encoder that obtains the best Spearman Correlation results in the STS 2016 (85.03).

[Ham and Kim \(2021\)](#) proposed both multilingual and monolingual models for measuring semantic similarity. The monolingual model uses a pre-trained XLM-R ([Conneau et al., 2020](#)) model as the basis. First, they train the model on two natural language inference datasets: SNLI ([Bowman et al., 2015](#)) and MLNI ([Williams et al., 2018](#)) for classification purposes. Then, they train the model on the STS Benchmark in an augmented way. To evaluate the model, they use the STS 2017 dataset, obtaining 89.80 Spearman Correlation.

5.4.1. Supervised $\times$ unsupervised

Among the selected approaches, there are supervised and unsupervised ones. Since this information was extracted from papers, we tried to visualize the results for each training method. First, [Fig. 11](#) illustrates the results obtained by supervised models in the selected datasets. The approach proposed by [Jiang et al. \(2022\)](#) (RoBERTa in the figure) is predominant among the results. It is followed by the approach by [Huang et al. \(2021\)](#) (BART in the figure). The work from [Arase and Tsujii \(2019\)](#) and the one from [MacLaughlin et al. \(2020\)](#) complete the figure. We describe the two most common approaches below.

[Jiang et al. \(2022\)](#) propose a new model called DCPCSE that improves an unsupervised BERT and a supervised RoBERTa. DCPCSE means the deep continuous prompt framework for contrastive learning of sentence embeddings. The authors freeze the language



Fig. 11. Best results of supervised models for Pearson and Spearman correlation in each dataset.

model layers, training only a deep continuous prompting layer that tries to generate the best prompting input for obtaining the expected output class. The modified RoBERTa supervised model obtained the best results compared to the unsupervised BERT. They evaluate using the Spearman correlation, obtaining: 79.14 in the STS 2012, 88.64 in the STS 2013, 83.73 in the STS 2014, 87.33 in the STS 2015, 84.57 in the STS 2016, 87.84 in the STS Benchmark, and 82.07 in the SICK-R dataset.

Huang et al. (2021) create a BART-based sentence embedding model that can disentangle semantics and syntax and perform better than the state-of-the-art in unsupervised semantic similarity tasks. The resulting ParaBART model works with two separate encoders, one for syntax and one for semantics. Their model manages to obtain the best result in STS datasets, reaching 80.90 in Pearson correlation in STS 2015.

Fig. 12 shows the results of the unsupervised models for the same set of datasets. Since the models in Fig. 10 combine both supervised and unsupervised and most of the best ones are unsupervised, both figures are similar. Thus, the work from Choi et al. (2021) is also the most common among the unsupervised approaches. Similarly, Jiang et al. (2019) also obtain the best results for the STS Benchmark. Li and Li (2024) propose an approach called AoE, which utilizes Transformer encoders, such as LLaMA (Touvron et al., 2023), to convert text into embeddings. In this method, the embedding from the last token serves as the sentence embedding. The AoE experiments involved LLaMA models with 7 billion (LLaMA7B) and 13 billion (LLaMA13B) parameters as the backbone. Additionally, the QLoRA technique was employed to fine-tune the LLaMA models. To obtain embeddings using LLaMA, the authors used the prompt: “Summarize sentence text in one word”, concatenating the resulting token to the last token of the text to generate the final embedding. Li and Li (2024) achieves the state-of-the-art of Spearman Correlation in four datasets with 79.3 on STS 2012, 90.7 on STS 2013, 90.5 on STS 2015 and 87.3 on STS 2016.

5.5. Datasets

About 60% of the datasets used in the classified papers come from SemEval, the International Workshop on Semantic Evaluation. Seven different versions focus on semantic textual similarity, called STS 2012–2017 and STS Benchmark. The others most used are SICK-Relatedness and MRPC (Microsoft Research Paraphrase Corpus).

Regarding on how the datasets determine the similarity between sentences, we have three distinct similarity evaluations: six-level, five-level, and a binary scale. All STS datasets use a six-level similarity, while SICK-R uses a five-level, and the MRPC uses a binary scale. Those datasets with six and five-level similarity scores are used for evaluation with Pearson and Spearman Correlation as metrics. In contrast, MRPC, which is binary, is often evaluated using accuracy, precision, recall, and F1-Score.

For us to understand these levels of similarity, we describe what they mean for the STS datasets. The six levels (0–5) of similarity defined by STS datasets consider the following criteria: 0 indicates that the pair of texts are completely different from each other; 1 means that two texts are not equivalent, but describe a related topic; 2 indicates pairs of texts that are not equivalent, but share some detailed definition; 3 and 4 mean that two texts are almost equivalent, however when 3, some important details are not the same, while 4 indicates that what is different between the texts is not important; 5 indicates that the two texts are semantically equivalent.

In order to better understand these datasets, we divide them according to their application domain. We defined eight distinct superclasses of domains. They are biomedical, machine translation, question answering, captions, forums, glosses, media



Fig. 12. Best results of unsupervised models for Pearson and Spearman correlation in each dataset.

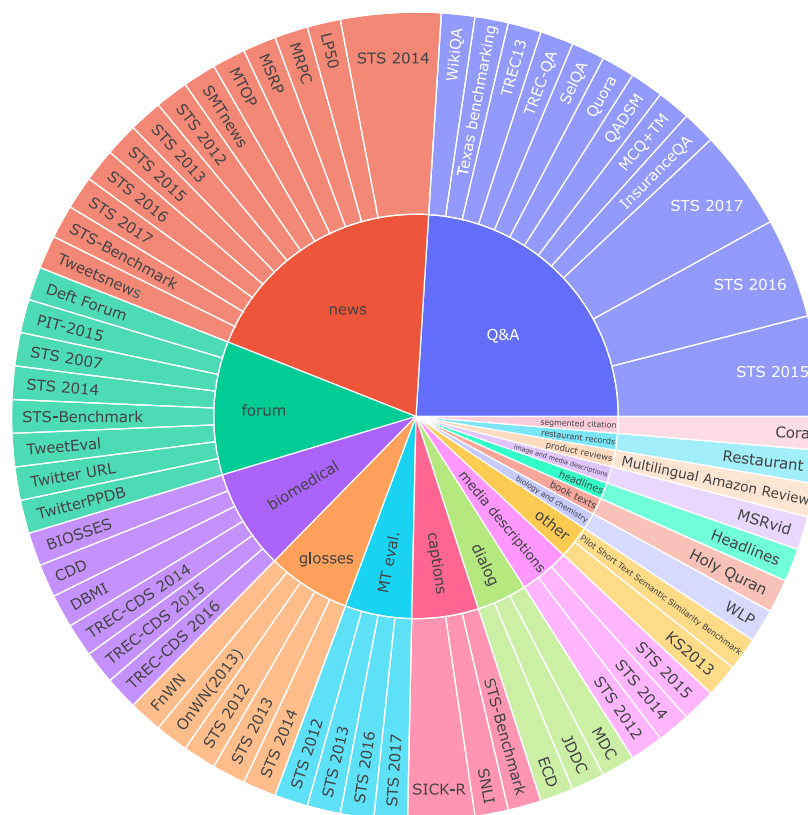


Fig. 13. Domain distribution of Semantic Textual Similarity evaluation datasets.

descriptions, and news. We detailed these super classes in Table 5 and Fig. 13. The most common domain is “news”. Eight datasets have at least part of their data from this domain. As Table 5 shows, we cluster multiple news-related topics from datasets and consider them as a “news” superclass.

Datasets can be a combination of multiple domains, which means that they can be used to evaluate models in different scenarios. However, in the biomedical context, we have datasets that only focus on this area, such as CDD, DBMI, TREC-QA, BIOSSES. Some of the domains can be a bit confusing, such as media descriptions and glosses. The type of texts contained in them are not properly described, so we cannot know properly what is the context of they.

One aspect that still needs improvement is the development of datasets for specific domains. Social networks, information technology, and medical datasets would stretch the capacity of existing models to new vocabularies. These domains would benefit from models that can deal with their textual structure.

5.6. Research guideline

From the data obtained in the studies and the analysis done, some patterns became clear. In this section, we summarize them in a guideline way. We describe what is the most common way to process sentences to obtain satisfactory results in the STS task.

- **Representing a Sentence:** When looking for ways to represent sentences, some models are preferred among the selected studies. BERT is the most popular model for generating sentence representations. Variations of BERT are also present in the list, such as RoBERTa and ALBERT. However, the one with the best results is SBERT (Reimers and Gurevych, 2019a). This model is a modification of BERT specifically to generate sentence representations. Approaches using it obtained some of the best results in the main datasets for the task.
- **Models for Semantic Similarity Classification:** From the analyzed studies, it became clear that the best models for semantic similarity computation are those that combine different existing models. The use of pre-trained models for generating sentence representations is predominant among the studies. Pre-trained models are trained on huge amounts of textual data. Therefore, they can capture linguistic patterns and convert them into text representations. BERT, SBERT, and RoBERTa are the most combined models. Surprisingly, CNN emerged as a useful model to process the output from these models. As we saw in our analysis, unsupervised models obtained the best results among all approaches. This means that focusing on generating the sentence representations can be more effective than training a model to calculate the similarity between two sentences.
- **Evaluation Dataset Selection:** There are several datasets with information about the semantic similarity between sentences. Some of the most used ones are those from the SemEval International Workshop. In order to evaluate a new approach to measure the semantic similarity between sentences, it is common to use STS 2012–2017, plus STS Benchmark. This allows the new approach to be compared against the state-of-the-art. Another common dataset is the SICK-R, which also comes from SemEval. This dataset contains annotations about the relatedness and entailment relationships between sentences.
- **Similarity Metrics:** The most common metric for similarity is Cosine. The majority of papers use Cosine to compare two-sentence representations and define whether they are similar or not. Existing work evaluates the use of other metrics, however, Cosine stands out as the main one.
- **Evaluation Metrics:** Although accuracy, precision, recall, and F1-score are used for some of the existing datasets (such as MRPC), the most common metrics are Pearson and Spearman correlation. These metrics are the main ones when using STS 2012–2017, and we can have a direct comparison against existing work. Pearson and Spearman correlation is used to compare the results obtained by a model with the gold standard scores from the datasets.

5.7. Open problems

Each identified approach brings a series of potential modifications that can be made to improve results. As we can see in Section 5.4, even the best approaches do not obtain higher than 0.95 for Pearson/Spearman correlation, as most of them can reach only 0.9 for some of the datasets. Here, we summarize some of the proposed modifications, and try to bring some ideas to improve models that generate meaningful sentence representations.

Considering the best approaches in terms of the presented results, we list some of their ideas for improvement. Choi et al. (2021), which propose the use of models such as ALBERT and BERT combined with convolutional layers, indicate that the larger version of these models can improve the resulting sentence representation. The use of other models can also improve the initial representation of words that are further processed. We saw that some approaches use transformer layers to process initial representations that will result in a sentence representation. Such a use of transformer layers can be employed in lieu of the suggested CNN by the authors.

Jiang et al. (2022) use pre-trained language models to generate word representations and manipulate the prompt input of transformers to obtain a richer sentence representation. The authors do not indicate any further modification to their word, however, it seems clear that the manipulation of the prompt input can be explored to obtain better sentence representations. In their work, they investigate only BERT and RoBERTa as pre-trained language models. Other modifications of BERT, such as ALBERT used by Choi et al. (2021), can provide representation that can fit better to the problem.

F. Liu et al. (2021), which propose TRANS-ENCODER, suggest that the improvement can come from the use of more diverse data, such as the Wikipedia corpus. They also suggest the use of a mix between lexical and semantic representations. The combination of lexical and semantics information from a sentence can bring extra elements for generating the representation. Lexical information can provide the importance of sentence elements considering the specific context they are.

The use of STS-related datasets allows us to evaluate the quality of text representations in terms of semantic similarity. Since these datasets are in English, it is important to expand to different languages and properly evaluate existing approaches to these languages. As other languages can have specific linguistic properties, approaches with good results for English may not have the same impact on them. Some existing multilingual approaches, such as [Cer et al. \(2018\)](#), [Yang et al. \(2020\)](#) and [Ham and Kim \(2021\)](#) obtain limited results in STS datasets compared to best approaches.

6. Conclusion

In this work, we presented a systematic literature review of sentence representation for the Semantic Textual Similarity Task. We defined a research protocol to obtain the most recent publications about the topic. As a result, we obtained a total of 108 papers from the obtained publications. From these papers, we read and extracted information for analysis and synthesis. We introduced a taxonomy for sentence representation in STS that divides the process into four main groups. Using the information from the obtained publications, we provided what are the main methods for each of the steps in the representation of sentences for STS. We tabulated and analyzed the extracted data, providing a series of graphs and visualizations that improved the understanding of the publications. From the obtained results, we identified the most used models for representing sentences and computing similarity. Based on the main datasets for STS, we were able to compare different approaches according to their results. Thus, we defined the best approaches for different datasets, which led us to state-of-the-art approaches. Throughout our analysis, some patterns arose, showing that combining pre-trained models with specific models for training can reach the best results. The use of Convolutional Neural Networks for representing sentences combined with BERT-based models can reach the best results in most existing datasets.

With the increasing number of existing approaches for representing texts, this systematic literature review provides to the research community with an overview of the STS task for sentences. In future work, we aim to use the knowledge obtained in this review to propose new combinations that can lead to new state-of-the-art approaches for sentence representation for semantic comparison.

CRedit authorship contribution statement

Larissa Guder: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **João Paulo Aires:** Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Hígor Uélinton da Silva:** Writing – review & editing, Methodology, Investigation. **Felipe Meneguzzi:** Writing – review & editing, Validation, Supervision, Investigation. **Dalvan Griebler:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dalvan Griebler reports financial support was provided by SAP SE. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Datasets details

See [Tables 4–6](#).

Appendix B. List of reviewed papers

See [Table 7](#).

Data availability

No data was used for the research described in the article.

Table 4

Best results in each dataset.

Dataset	Approach	Pearson	Spearman	Approach	Pearson	Spearman
	Supervised			Unsupervised		
STS 2012	BART (Huang et al., 2021)	0.684	0.791	Statistical Model (Yu et al., 2020a)	0.760	0.793
	RoBERTa (Jiang et al., 2022)			LLAMA13b (Li and Li, 2024)		
STS 2013	BART (Huang et al., 2021)	0.711	0.886	CNN-SBERT (Choi et al., 2021)	0.899	0.907
	RoBERTa (Jiang et al., 2022)			LLAMA13b (Li and Li, 2024)		
STS 2014	BART (Huang et al., 2021)	0.764	0.837	CNN-SBERT (Choi et al., 2021)	0.905	0.892
	RoBERTa (Jiang et al., 2022)			CNN-SBERT (Choi et al., 2021)		
STS 2015	BART (Huang et al., 2021)	0.809	0.873	CNN-SBERT (Choi et al., 2021)	0.894	0.905
	RoBERTa (Jiang et al., 2022)			LLAMA13b (Li and Li, 2024)		
STS 2016	BART (Huang et al., 2021)	0.801	0.846	Statistical Model (Yu et al., 2020a)	0.862	0.873
	RoBERTa (Jiang et al., 2022)			LLAMA13b (Li and Li, 2024)		
STS 2017				BERT (Atabuzzaman et al., 2021b)	0.812	0.898
				XL-M-R (Ham and Kim, 2021)		
STS Benchmark	Custom (Y. Yu et al., 2024)	0.884	0.878	RoBERTa (Jiang et al., 2019)	0.929	0.925
	RoBERTa (Jiang et al., 2022)			RoBERTa (Jiang et al., 2019)		
SICK	RandomSubSpace (MacLaughlin et al., 2020)	0.978	0.970	SBERT (Wang and Kuo, 2020)	0.876	0.829
	RandomSubSpace (MacLaughlin et al., 2020)			SBERT (Wang and Kuo, 2020)		

Table 5

Datasets domain.

Group	Source	Dataset
biomedical	biomedical	CDD, DBMI, TREC-QA, BIOSSES
MT eval.	MT eval.	STS 2013
	MT posteds	STS 2016, STS 2017
	WMT eval.	STS 2012
Q&A	Q&A forum answers	STS 2015, STS 2016, STS 2017
	Q&A forum questions	STS 2016, STS 2017
	Questions pairs	Quora
	committed belief	STS 2015
	short-answer plag.	STS 2016, STS 2017
	student answers	STS 2015
captions	captions	STS-Benchmark
	image caption	SICK-R
	video caption	SICK-R
forum	forum posts	STS 2014, STS-Benchmark
	twitter	PIT-2015
glosses	glosses	STS 2012, STS 2013, STS 2014
media descriptions	image descriptions	STS 2014, STS 2015
	videos	STS 2012
news	news	STS-Benchmark, MSRP
	news summary	STS 2014
	newswire	STS 2012, STS 2013
	newswire headlines	STS 2014, STS 2015, STS 2016, STS 2017
	tweet-news pairs	STS 2014

Table 6
Datasets structure.

Dataset	Composition	Size	Similarity range	Metrics
STS-Benchmark (Cer et al., 2017)	MSRpar	8,628	0–5	Spearman/Pearson
	headlines	1500		
	deft-news	2499		
	MSRvid	300		
	images	1500		
	track5.en-en	1500		
	deft-forum	250		
	ans-forums	450		
	ans-ans	375		
STS 2012 (Agirre et al., 2012)		254	0–5	Spearman/Pearson
	MSRpar	1500		
	MSRvid	1500		
	OnWN	750		
	SMTnews	750		
	SMTeuroparl	750		
STS 2013 (Agirre et al., 2013)		2250	0–5	Spearman/Pearson
	HDL	750		
	FNWN	189		
	OnWN	561		
	SMT	750		
STS 2014 (Agirre et al., 2014)		3750	0–5	Spearman/Pearson
	HDL	750		
	OnWN	750		
	Deft-forum	450		
	Deft-news	300		
	Images	750		
	Tweet-news	750		
STS 2015 (Agirre et al., 2015)		3000	0–5	Spearman/Pearson
	HDL	750		
	Images	750		
	Ans.-student	750		
	Ans.-forum	375		
	Belief	375		
STS 2016 (Agirre et al., 2016)			0–5	Spearman/Pearson
	HDL	249		
	Plagiarism	230		
	post-editing	244		
	Ans.-Ans.	254		
	Quest.-Quest	209		
SICK-R (Marelli et al., 2014)	Image Captions	9840	1–5	Spearman/Pearson
MRPC (Dolan and Brockett, 2005)	News	5801	0–1	Accuracy/F1/Pearson

Table 7
Covered papers on the review.

Ref	Title	Year
Y. Yu et al. (2024)	A Quantum-Like Tensor Compression Sentence Representation Based on Constraint Functions for Semantics Analysis	2024
Li and Li (2024)	AoE: Angle-optimized Embeddings for Semantic Textual Similarity	2024
Huang et al. (2024)	CoSENT: Consistent Sentence Embedding via Similarity Ranking	2024
C. Yu et al. (2024)	Enhancing Sentence Representation with a Transfer Task Enhancement Auxiliary Network	2024
Feng et al. (2024)	Focusing on differences! Sample framework enhances semantic textual similarity with external knowledge	2024
Wang and Lv (2024)	Prefix Data Augmentation for Contrastive Learning of Unsupervised Sentence Embedding	2024
Salama et al. (2024)	Semantic Compression for Word and Sentence Embeddings using Discrete Wavelet Transform	2024
Kang et al. (2024)	Semantic Similarity Analysis via Syntax Dependency Structure and Gate Recurrent Unit	2024

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Table 7 (continued).

Ref	Title	Year
Zhou et al. (2024)	Sentence Similarity Model based on Semantics and Syntax	2024
Wang et al. (2024)	VAEGPT-Sim: Improving Sentence Representation with Limited Corpus Using Gradually-Denoising VAE	2024
Ma et al. (2023)	A Contrastive Framework to Enhance Unsupervised Sentence Representation Learning	2023
Zhou et al. (2023)	A Short-Text Similarity Model Combining Semantic and Syntactic Information	2023
Miao et al. (2023)	DebCSE: Rethinking Unsupervised Contrastive Sentence Embedding Learning in the Debiasing Perspective	2023
Hany et al. (2023)	Enhancing Textual Relatedness Assessment with Combined Transformers-Embedding Similarity Techniques and Machine Learning Regressors	2023
J. Liu et al. (2023)	ESCL: Equivariant Self-Contrastive Learning for Sentence Representations	2023
Xu et al. (2023)	HSimCSE: Improving Contrastive Learning of Unsupervised Sentence Representation with Adversarial Hard Positives and Dual Hard Negatives	2023
T. Liu et al. (2023)	Parameter-Effective Contrastive Learning and Isotropy of Different Layers For Better Sentence Embeddings	2023
Li et al. (2023)	Siamese BERT Architecture Model with attention mechanism for Textual Semantic Similarity	2023
Dang et al. (2023)	SubTST: a consolidation of sub-word latent topics and sentence transformer in semantic representation	2023
Wang et al. (2023)	Unsupervised Sentence Embedding Model Based on Contrastive Learning	2023
Zhang et al. (2022)	A Contrastive Framework for Learning Sentence Representations from Pairwise and Triple-wise Perspective in Angular Space	2022
Tan et al. (2022)	A Sentence is Worth 128 Pseudo Tokens: A Semantic-Aware Contrastive Learning Framework for Sentence Embeddings	2022
Saeed and Goma (2022)	An Ensemble-Based Model to Improve the Accuracy of Automatic Short Answer Grading	2022
Zhao et al. (2022)	Compressing Sentence Representation for Semantic Retrieval via Homomorphic Projective Distillation	2022
Wolfe and Caliskan (2022)	Contrastive Visual Semantic Pretraining Magnifies the Semantics of Natural Language Representations	2022
Jiang et al. (2022)	Deep Continuous Prompt for Contrastive Learning of Sentence Embeddings	2022
Feng et al. (2022)	Language-agnostic BERT Sentence Embedding	2022
Meenakshi and Shanavas (2022)	Novel Shared Input Based LSTM for Semantic Similarity Prediction	2022
Wieting et al. (2022)	Paraphrastic Representations at Scale	2022
Ni et al. (2022)	Sentence-T5: Scalable Sentence Encoders from Pre-trained Text-to-Text Models	2022
Xie et al. (2022)	Stable Contrastive Learning for Self-Supervised Sentence Embeddings With Pseudo-Siamese Mutual Learning	2022
Seo et al. (2022)	TA-SBERT: Token Attention Sentence-BERT for Improving Sentence Representation	2022
Sonbol et al. (2020)	Text similarity calculation method of Siamese network based on ALBERT	2022
Liu et al. (2022)	TRANS-ENCODER: UNSUPERVISED SENTENCE-PAIR MODELLING THROUGH SELF- AND MUTUAL-DISTILLATIONS	2022
L. Wang et al. (2021)	Aligning Cross-lingual Sentence Representations with Dual Momentum Contrast	2021
Yu et al. (2021)	An efficient framework for sentence similarity inspired by quantum computing	2021
Lan and He (2021)	Answer Selection Based on Aligned Local Composite Features and Global Features	2021
Li et al. (2021)	Biomedical Text Similarity Evaluation Using Attention Mechanism and Siamese Neural Network	2021
Y. Zhang et al. (2021)	Bootstrapped Unsupervised Sentence Representation Learning	2021
Wang et al. (2021b)	Comparison between Calculation Methods for Semantic Text Similarity Based on Siamese Networks	2021
Yan et al. (2021)	ConSERT: A Contrastive Framework for Self-Supervised Sentence Representation Transfer	2021
Goswami et al. (2021b)	Cross-lingual Sentence Embedding using Multi-Task Learning	2021
Tsukagoshi et al. (2021)	DefSent: Sentence Embeddings using Definition Sentences	2021
C. Liu et al. (2021)	DialogueCSE: Dialogue-based Contrastive Learning of Sentence Embeddings	2021
Huang et al. (2021)	Disentangling Semantics and Syntax in Sentence Embeddings with Pre-trained Language Models	2021
Choi et al. (2021)	Evaluation of BERT and ALBERT Sentence Embedding Performance on Downstream NLP Tasks	2021
Di Giovanni and Brambilla (2021)	Exploiting Twitter as Source of Large Corpora of Weakly Similar Pairs for Semantic Sentence Embeddings	2021
Sun et al. (2021)	GCNs-Based Context-Aware Short Text Similarity Model	2021
Liang and Zhang (2021)	Generating and Measuring Similar Sentences Using Long Short-Term Memory and Generative Adversarial Networks	2021
Tiyajamorn et al. (2021)	Language-agnostic Representation from Multilingual Sentence Encoders for Cross-lingual Similarity Estimation	2021
Atabuzzaman et al. (2021a)	Leveraging Grammatical Roles for Measuring Semantic Similarity Between Texts	2021
Min et al. (2021)	Locality Preserving Sentence Encoding	2021
Goswami et al. (2021a)	Mufin: Enriching Semantic Understanding of Sentence Embedding using Dual Tune Framework	2021
Prabhu et al. (2021)	On-Device Sentence Similarity for SMS Dataset	2021
D. Zhang et al. (2021)	Pairwise Supervised Contrastive Learning of Sentence Representations	2021
Masлак and Mitkov (2021)	Paragraph Similarity Matches for Generating Multiple-choice Test Items	2021
Jin et al. (2021)	Promotion of Answer Value Measurement With Domain Effects in Community Question Answering Systems	2021
Singh et al. (2021)	Question Answering Chatbot using Deep Learning with NLP	2021

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Table 7 (continued).

Ref	Title	Year
Ham and Kim (2021)	Semantic Alignment with Calibrated Similarity for Multilingual Sentence Embedding	2021
Atabuzzaman et al. (2021b)	Semantic Representation of Sentences Employing an Automated Threshold	2021
Zhu et al. (2021)	Siamese ELECTRA Network Combined with BERT for Semantic Similarity	2021
Gao et al. (2021)	SimCSE: Simple Contrastive Learning of Sentence Embeddings	2021
Jiang et al. (2019)	SMART: Robust and Efficient Fine-Tuning for Pre-trained Natural Language Models through Principled Regularized Optimization	2021
Peng et al. (2021)	Structure-aware Sentence Encoder in Bert-Based Siamese Network	2021
Wang et al. (2021a)	TSDAE: Using Transformer-based Sequential Denoising Auto-Encoder for Unsupervised Sentence Embedding Learning	2021
Ha et al. (2021)	Utilizing SBERT For Finding Similar Questions in Community Question Answering	2021
Wieting et al. (2020)	A Bilingual Generative Transformer for Semantic Sentence Embedding	2020
Yu et al. (2020a)	A Quantum Entanglement-Based Approach for Computing Sentence Similarity	2020
Zhong et al. (2020)	A Semi-Supervised Feature Fusion Method for Sentence Similarity Matching	2020
Swarup et al. (2020)	An Instance Level Approach for Shallow Semantic Parsing in Scientific Procedural Text	2020
Zhang et al. (2020)	An Unsupervised Sentence Embedding Method by Mutual Information Maximization	2020
Li et al. (2020)	Combining Local and Global Features Into a Siamese Network for Sentence Similarity	2020
Z. Li et al. (2020a)	Cross2Self-attentive Bidirectional Recurrent Neural Network with BERT for Biomedical Semantic Text Similarity	2020
Lattar et al. (2020)	Duplicate record detection approach based on sentence embeddings	2020
B. Wang et al. (2021)	Efficient sentence embedding via semantic subspace analysis	2020
MacLaughlin et al. (2020)	Evaluating the Effectiveness of Efficient Neural Architecture Search for Sentence-Pair Tasks	2020
Li and Wu (2020)	Exploiting WordNet Synset and Hypernym Representations for Answer Selection	2020
Shin et al. (2020)	Fast and Accurate Deep Bidirectional Language Representations for Unsupervised Learning	2020
Xiao et al. (2020)	HINDBR: Heterogeneous Information Network Based Duplicate Bug Report Prediction	2020
Li et al. (2020b)	Interactive Self-Attentive Siamese Network for Biomedical Sentence Similarity	2020
Brychcín (2018)	Linear transformations for cross-lingual semantic textual similarity	2020
Ahmed et al. (2020)	Multilingual Semantic Textual Similarity using Multilingual Word Representations	2020
Saeed et al. (2020)	On Finding Similar Verses from the Holy Quran using Word Embeddings	2020
B. Li et al. (2020a)	On the Sentence Embeddings from Pre-trained Language Models	2020
Yu et al. (2020b)	Quantum Entanglement Based Sentence Similarity Computation	2020
Wang and Kuo (2020)	SBERT-WK: A Sentence Embedding Method by Dissecting BERT-Based Word Models	2020
Besti et al. (2020)	Semantic Classification of Scientific Sentence Pair Using Recurrent Neural Network	2020
Zhou et al. (2020)	Sentence Modeling via Graph Construction and Graph Neural Networks for Semantic Textual Similarity	2020
Chen et al. (2020)	Sentence Pair Similarity Modeling Based on Weighted Interaction of Multi-semantic Embedding Matrix	2020
Peinelt et al. (2020)	tBERT: Topic Models and BERT Joining Forces for Semantic Similarity Detection	2020
Tang et al. (2020)	Text semantic understanding based on knowledge enhancement and multi-granular feature extraction	2020
Sonbol et al. (2020)	Towards a Semantic Representation for Functional Software Requirements	2020
Fulda (2020)	You Are What You Read: The Effect of Corpus and Training Task on Semantic Absorption in Recurrent Neural Architectures	2020
M. Chen et al. (2019)	A Multi-Task Approach for Disentangling Syntax and Semantics in Sentence Representations	2019
Quan et al. (2019)	An Efficient Framework for Sentence Similarity Modeling	2019
Sileo et al. (2019)	Composition of Sentence Embeddings: Lessons from Statistical Relational Learning	2019
Zhang et al. (2019)	Disease Prediction and Early Intervention System Based on Symptom Similarity Analysis	2019
Wijnholds and Sadrzadeh (2019)	Evaluating Composition Models for Verb Phrase Elliptical Sentence Embeddings	2019
Tang et al. (2019)	Fusing Syntax and Word Embedding Knowledge for Measuring Semantic Similarity	2019
D. Liu et al. (2019)	Multi-Scale Deformable CNN for Answer Selection	2019
Blagec et al. (2021)	Neural sentence embedding models for semantic similarity estimation in the biomedical domain	2019
S. Chen et al. (2019)	Self-Attention based Network For Medical Query Expansion	2019
Reimers and Gurevych (2019b)	Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks	2019
Carta and Bacciu (2019)	Sequential Sentence Embeddings for Semantic Similarity	2019

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Table 7 (continued).

Ref	Title	Year
Peng et al. (2019)	Short Text Clustering Enhanced by Semantic Matching Model	2019
Arase and Tsujii (2019)	Transfer Fine-Tuning: A BERT Case Study	2019
Garí Soler et al. (2019)	Word Usage Similarity Estimation with Sentence Representations and Automatic Substitutes	2019
Bao et al. (2018)	Attentive Siamese LSTM Network for Semantic Textual Similarity Measure	2018
Lan and Xu (2018)	Character-Based Neural Networks for Sentence Pair Modeling	2018
Tymoshenko and Moschitti (2018)	Cross-Pair Text Representations for Answer Sentence Selection	2018
Gong et al. (2018)	Deep Variation Autoencoder with Topic Information for Text Similarity	2018
Luo et al. (2018)	Integrating Deep Learning Approaches for Identifying News Reprint Relation	2018
Samardzhiev et al. (2018)	Learning Neural Word Saliency Scores	2018
Yang et al. (2018)	Learning Semantic Textual Similarity from Conversations	2018
Ding and Xu (2018)	Learning Sentence Embeddings Based on Weighted Contexts from Unlabeled Data	2018
Ichida et al. (2018)	Measuring Semantic Similarity Between Sentences Using A Siamese Neural Network	2018
Wang et al. (2018)	Paraphrase Recognition via Combination of Neural Classifier and Keywords	2018
Chali and Islam (2018)	Question-Question Similarity in Online Forums	2018
Farouk (2018)	Sentence Semantic Similarity based on Word Embedding and WordNet	2018
Wu et al. (2018)	Word Mover's Embedding: From Word2Vec to Document Embedding	2018

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